

Science
Grade 11, Chemistry
SCH3U
Final Examination
ANSWER KEY

Name: _____ Date _____

Instructions:

- 1) **You will have the entire period to complete the test (90 mins)**
- 2) **The test is made up of 2 sections:**
 - **Section A – Multiple Choice / True and False**
 - **Section B – Long Answer**
- 3) **Please follow all the instructions**
- 4) **Please read all questions carefully**
- 5) **A mark breakdown of the test can be found after these instructions**
- 6) **Good Luck!**

Mark Summary

Section	Knowledge /Understanding	Thinking	Application	Communication	TOTAL
A					40
B					50
Sub-Totals					/ 90

Comments:

Part A: Multiple Choice [40 questions / 40 marks /]

- If a chemical substance enters your eye, you should:
 - wait to tell the teacher, then go to the eyewash station and rinse your eyes for 15 minutes
 - go immediately to the eyewash station, hold your eyelids back with your fingers, and rinse your eyes for 15 minutes; have someone inform the teacher as soon as possible
 - run water into the nearest sink and splash water on your face immediately
 - rub your eyes vigorously to start tears flowing to flush out the chemical
- Which one of the following is a metalloid?
 - Rh, Rhodium (element # 45)
 - Po, Polonium (element # 84)
 - Sr, Strontium (element # 38)
 - Ar, Argon (element # 18)
- Which compound is an acid?
 - HCl
 - NaCl
 - Ca(OH)₂
 - CH₄
- Which compound would exhibit covalent bonding?
 - Fe₂O₃
 - MgCl₂
 - NF₃
 - KBr
- Which element would be the **worst** conductor of electricity?
 - Cr, chromium (element # 24)
 - Ir, iridium (element # 77)
 - Zr, zirconium (element # 40)
 - Kr, krypton (element # 36)
- The group whose elements have the greatest electron affinity is
 - 1
 - 2
 - 17
 - 18
- A box of baking powder carries the statement: "Ingredients: corn starch, sodium bicarbonate, calcium hydrogen phosphate, and sodium sulfate." Baking powder is best described as:
 - a pure substance
 - a mixture of compounds
 - a solution
 - a mixture of elements
- When the following chemical equation is correctly balanced,



the sum of the coefficients is:

- A) 7 B) 8 C) 9 **D) 19**

9. Determine the number of iron atoms in a 55.85 g sample of iron metal.

- A) 6.02×10^{23}** B) 2.05×10^{23} C) 0.343 D) 3.68×10^{23}

10. How many **protons** / **electrons** / **neutrons** are in the isotope tin-119?

- A) 50 / 50 / 69** B) 50 / 50 / 19 C) 69 / 50 / 50 D) 59 / 59 / 1

11. All of the following balanced nuclear reactions are correctly classified except:

- A) $^{235}_{92}\text{U} + ^1_0\text{n} \rightarrow ^{94}_{38}\text{Sr} + ^{140}_{54}\text{Xe} + 2^1_0\text{n}$ fission
B) $^2_1\text{H} + ^3_1\text{H} \rightarrow ^4_2\text{He} + ^1_0\text{n}$ alpha decay
C) $^{28}_{12}\text{Mg} \rightarrow ^0_{-1}\text{e} + ^{28}_{13}\text{Al}$ beta decay
D) $^{27}_{13}\text{Al} + ^4_2\text{He} \rightarrow ^{31}_{15}\text{P} + ^4_2\text{He}$ fusion

12. Which salt is insoluble in water?

- A) NaOH B) $\text{Cr}_2(\text{SO}_4)_3$
C) CdS D) AgNO_3

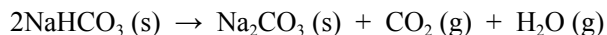
13. All of the following are properties of molecular compounds except:

- A) low melting and boiling points
B) low solubility in water
C) poor conductivity
D) crystalline structure

14. Which of the following general reactions is incorrect?

- A) $\text{ACID} + \text{BASE} \rightarrow \text{WATER} + \text{SALT}$
B) $\text{METAL} + \text{HALOGEN} \rightarrow \text{METAL HALIDE}$
C) $\text{METAL OXIDE} + \text{WATER} \rightarrow \text{ACID}$
D) $\text{ACID} + \text{METAL} \rightarrow \text{SALT} + \text{HYDROGEN}$

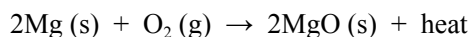
15. Classify the following balanced chemical reaction:



- A) Combination
B) Decomposition

- C) Combustion
- D) Single Displacement

16. Which classification applies to the following balanced chemical reaction?



- A) Combustion
- B) Combination
- C) Exothermic
- D) all of the above

17. The absence of iodine in our diet results in an abnormal enlargement of the thyroid gland. The following **unbalanced** chemical equation describes the laboratory preparation of iodine. Balance the equation and determine the reacting mole ratio of $n_{\text{H}_2\text{SO}_4} / n_{\text{KI}}$.



The reacting mole ratio of $n_{\text{H}_2\text{SO}_4} / n_{\text{KI}}$ is:

- A) 1.5
- B) 0.67
- C) 1
- D) 0.5

18. Given 10.0 g samples of each of the following substances, which contains the greatest number of moles?

- A) CO_2 (M = 44.01)
- B) CCl_4 (M = 153.81)
- C) C_3H_8 (M = 44.11)
- D) CH_3F (M = 34.04)

19. Which of the following would have the lowest density at room temperature?

- A) Francium (Fr)
- B) Tin (Sn)
- C) Radon (Rn)
- D) Mercury (Hg)

20. Which of the following has the smallest atomic radius?

- A) Ge
- B) As
- C) Se
- D) Br

21. The periodic table is organized by increasing atomic number. With only five exceptions (Ar < K, for example) the table is also organized by increasing atomic mass. Which of the following explanations is most likely to account for these anomalies?

- A) Experimental error.
- B) The different masses of ions of Ar.
- C) Fewer numbers of protons in K than Ar.
- D) The number of neutrons in an average atom of these elements.

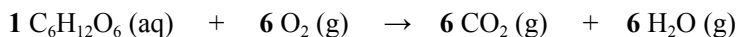
22. All of the following are equivalent to one mole except

- A) 6.02×10^{23} atoms of Ne
- B) exactly 12 g of carbon
- C) 18.0 g of water
- D) All of the above

23. For which molecule is it impossible to draw a Lewis dot structure that obeys the Octet rule?

- A) CO_2
- B) CHCl_3
- C) NO
- D) NH_3

24. For the **balanced** reaction between glucose and oxygen (aka respiration):



Choose the most accurate statement:

- A) The reaction always produces equal amounts of carbon dioxide and water molecules.
- B) 1.0 mol of glucose and 1.0 mol of oxygen will produce 1.0 mol of carbon dioxide.
- C) 10 g of glucose will need 60 g of oxygen to react completely.
- D) 1 atom of glucose reacts with 6 atoms of O_2 to produce 6 atoms each of CO_2 and H_2O .

25. Which ion is isoelectronic with Kr?

- A) As^{3-}
- B) Sr^{2-}
- C) Kr
- D) Br^+

26. 6.02×10^{23} molecules is to one mole as

- A) nomenclature is to gnomes
- B) Colonel Sanders is to chicken
- C) 18 days are to one winter holiday
- D) 12 donuts are to one dozen donuts

27. Which of the following elements has the highest electronegativity?

- A) O
- B) Ne
- C) Be
- D) S

28. Which bond is most polar (greatest difference in electronegativities)?

- A) F – F
- B) C – N
- C) C – O
- D) N – N

29. Which of the following elements has the lowest reactivity with water?

A) Ca

B) Mg

C) K

D) Na

30. Which compound would be most ionic in character?

A) NaCl

B) NaBr

C) RbI

D) RbF

31. Which is the correct formula for cobalt (III) iodate (V)?

A) $\text{Co}(\text{IO}_3)_3$

B) $\text{Co}_3(\text{IO}_3)_5$

C) $\text{Co}_5(\text{IO}_5)_3$

D) $\text{Co}(\text{IO}_5)_3$

32. In which of the following does chromium have the highest valence or oxidation state?

A) Cr_2O_3

B) CrCl_2

C) $\text{Na}_2\text{Cr}_2\text{O}_7$

D) $\text{Al}_2(\text{CrO}_3)_3$

33. Which element has the highest ionization energy?

A) Lithium

B) Carbon

C) Neon

D) Iodine

34. The reactivity of alkali metals in water to form hydrogen gas and metal hydroxide solutions increases down the group. This trend is due, in part, to:

A) decreasing ionization energies.

B) increasing electronegativities.

C) increasing reactivity of water.

D) increasing numbers of neutrons in the nucleus.

35. The double displacement reaction between an **acid** and a **base** is best classified as:

A) precipitation

B) neutralization

C) gas formation

D) all of the above

36. If there are 2.0 mol of O atoms in Fe_2O_3 , how many moles of iron atoms are also present?

A) 4/3

B) 3.0

C) 2/3

D) 3/2

37. How many chlorine atoms are in 7.45 mol of CCl_4 ($M_{\text{CCl}_4} = 153.81$)

A) $7.45 \times 6.02 \times 10^{23}$

B) 7.45×4

C) $7.45 \times 4 \times 6.02 \times 10^{23}$

D) $7.45 \times 4 \times 6.02 \times 10^{23} \times 153.81$

38. If XF_2 is the correct formula for a metallic fluoride, then the formula for the nitride of X is

A) X_3N

B) X_3N_2

C) X_2N_3

D) XN_2

39. Which substance is ionic?

Substance	Melting Point (°C)	Boiling Point (°C)	Conductivity
<i>A</i>	−80	−20	none
<i>B</i>	20	190	none
<i>C</i>	320	770	as solid
<i>D</i>	800	1250	in solution

A) *A*

B) *B*

C) *C*

D) *D*

40. Which molecule is non-polar?

A) H₂O

B) CO₂

C) NH₃

D) CO

Part B: Long Answers [10 questions / 50 marks]

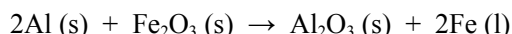
Step 2: Determine the mole ratio of Os to F₂: Divide the number of moles of Os by the number of moles of F₂: Mole ratio of Os to F₂ = (0.00999 moles Os) / (0.0300 moles F₂) = 1/3

Step 3: Write the empirical formula: Since the mole ratio of Os to F₂ is 1/3, this means that the compound has one osmium atom for every three fluorine atoms.

The empirical formula will be OsF₃, indicating that there is one osmium atom (Os) for every three fluorine atoms (F).

So, the chemical formula of the product is OsF₃.

5. The “thermite” reaction is a spectacular single displacement reaction between aluminum and iron (III) oxide. Consider the balanced chemical equation for the reaction:



- A) What evidence is there that this reaction is highly exothermic? [2 marks]

Heat and Light Production, Spontaneous Ignition, Temperature Rise

- B) If 104.5 g of aluminum and 559.4 g Fe₂O₃ (M_{Fe₂O₃} = 159.70) are mixed, which reactant is limiting? Show all of your work for full marks. [3 marks]

Number of moles of Al = 104.5 g / 26.98 g/mol ≈ 3.878 moles

Molar mass of Fe₂O₃ = 2 * 55.85 g/mol + 3 * 16.00 g/mol = 111.70 g/mol

Number of moles of Fe₂O₃ = 559.4 g / 111.70 g/mol ≈ 5.011 moles

From 3.878 moles of Al, you can form (3.878 moles) / 2 = 1.939 moles of Al₂O₃.

From 5.011 moles of Fe₂O₃, you can form (5.011 moles) / 1 = 5.011 moles of Al₂O₃.

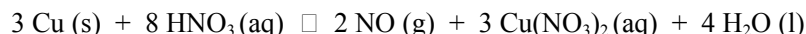
Since aluminum can produce fewer moles of Al₂O₃ compared to iron(III) oxide, it is the limiting reactant in this reaction.

6. A) Determine the number of each subatomic particle in the phosphide ion. [3 marks]

	# of protons	# of electrons	# of neutrons
$^{31}_{15}\text{P}^{3-}$	15	12	16

- B) Draw the Bohr-Rutherford atomic model of the phosphide ion, $^{31}_{15}\text{P}^{3-}$. Show clearly the relative positions of the subatomic particles in the ion. [3 marks]

One laboratory method used to generate nitrogen (II) oxide uses the reaction of copper metal with aqueous nitric acid. The balanced chemical equation for the reaction is:



A) Calculate the amount of copper in a 25.0 g sample of the pure metal. [2 marks]

Number of moles of Cu = $25.0 \text{ g} / 63.55 \text{ g/mol} \approx 0.394 \text{ moles}$

B) Calculate the maximum mass, in grams, of NO (g) that could be produced in a reaction between 25.0 g of Cu and excess HNO₃. [4 marks]

Number of moles of Cu = $25.0 \text{ g} / 63.55 \text{ g/mol} \approx 0.394 \text{ moles}$

$(0.394 \text{ moles Cu}) \times (2 \text{ moles NO} / 3 \text{ moles Cu}) = 0.263 \text{ moles NO}$

Mass of NO (g) = $0.263 \text{ moles} \times (30.01 \text{ g/mol}) \approx 7.90 \text{ g}$

Therefore, the maximum mass of NO (nitrogen(II) oxide) that could be produced in the reaction is approximately 7.90 grams when 25.0 g of copper is used with excess HNO₃.

In an experiment, 5.00 g of silver (I) nitrate (M = 169.88) is added to an aqueous solution containing an excess of sodium bromide (M = 102.89). It was found that 5.03g of solid silver (I) bromide (M = 187.77) precipitated in this double displacement reaction.

A) Determine the theoretical yield of AgBr. [4 marks]

Number of moles of AgNO₃ = $5.00 \text{ g} / 169.88 \text{ g/mol} \approx 0.02943 \text{ moles}$

Number of moles of AgBr = $5.00 \text{ g} / 169.88 \text{ g/mol} \approx 0.02943 \text{ moles}$

Mass of AgBr = $0.02943 \text{ moles} \times 187.77 \text{ g/mol} \approx 5.52 \text{ g}$

So, the theoretical yield of AgBr is approximately 5.52 grams

B) Calculate the percentage yield of the experiment. [2 marks]

Percent Yield =

$\frac{5.52 \text{ g}}{5.03 \text{ g}}$

$\times 100\% \approx 91.0\%$

7. Consider the following ions: S²⁻, Cl⁻, and Ca²⁺. Use your understanding of atomic structure explain which ion has the smallest radius. [3 marks]

Ca²⁺ (calcium ion):

Calcium has lost two electrons to become positively charged.

It has the same number of protons as a neutral calcium atom (Z = 20), but it has two fewer electrons.

The loss of electrons reduces the electron-electron repulsion, allowing the remaining electrons to be held closer to the nucleus by the stronger positive charge.

As a result, the calcium ion is smaller than a neutral calcium atom.

8.

Marking Scheme

Balanced Eqn /1

Mole Calculation /1

Mole ratio /1

Mass Calculation /1

Radioactivity at home

You may be surprised to learn that you do not need to visit a nuclear power plant or a hospital X-ray laboratory to find sources of radioactivity. They are all around us. In fact, it is likely that you'll find a few at home. Your front porch may incorporate cinder blocks or granite blocks. Both contain uranium. Walk through the front door, look up, and you'll see a smoke detector that owes its effectiveness to the constant source of alpha particle emissions from Americium-241. As long as the gases remain ionized within the shielded container, electricity flows, and all is calm. When smoke enters the chamber, it neutralizes the charges on these ions. In the absence of these ions, the circuit breaks and the alarm goes off.

Indicator lights on your appliances may use Krypton-85; electric blankets, promethium-147; and fluorescent lights, thorium-229. Even the food we eat is radioactive. The more potassium-rich the food source, the more potassium-40—a radioactive isotope that makes up about 0.01% of the natural supply of this mineral—is present. Thus, brazil nuts, peanuts, bananas, potatoes, and flour, all rich in potassium, are radiation sources.

—*Chem Matters*
April 2000

- A) Write the equation for the alpha decay that occurs in a smoke detector containing Americium-241. Gamma radiation is released. [4 marks]



- B) Explain why the release of gamma radiation usually accompanies any change to the nucleus of an atom. [2 marks]

The release of gamma radiation accompanies changes to the nucleus of an atom because it is a natural consequence of nuclear processes that involve energy changes and transitions in the nucleus. Gamma radiation allows the nucleus to release excess energy and attain a more stable and lower energy state.

- C) The complete decay of ${}_{95}^{241}\text{Am}$ involves successive α , α , β , α , β , α , α , β , α , and β emission. What is the final stable nucleus produced in this decay series? [1 mark]